

MARCOS AGUILERA-KEYSER

APPLIED SCIENCE | GENERATIVE AI, RETRIEVAL & EVALUATION

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Applied ML professional combining scientific investigation with hands-on implementation across enterprise analytics and insurance. Since 2023, work has spanned production RAG and agentic QA systems, followed by autonomous applied research in a Generative AI Center of Excellence. Focus areas include performance diagnosis, model adaptation, retrieval, and evaluation of complex procedural text.

RESEARCH & TECHNICAL FOCUS

Experimental design and failure analysis; RAG evaluation; domain adaptation of encoders, cross-encoders, and LLMs; neural and knowledge-graph retrieval; reinforcement learning with verifiable rewards.

PROFESSIONAL EXPERIENCE

Principal Applied Research Scientist - Generative AI

New York Life | Jan 2023 - Jul 2026

- Investigated performance limitations in production RAG QA over insurance procedures, combining quantitative corpus and semantic-drift analysis with qualitative review of questions, retrieved evidence, and answers.
- Conducted domain-adaptation work across bi-encoders, cross-encoders, and generative models, including masked-language-model training for MiniLM and QLoRA adaptation of Llama 3.1 8B.
- Developed retrieval experiments and evaluation workflows covering lexical and dense baselines, relevance judgments, data-split controls, and model comparisons.
- Designed and implemented a procedural-RAG prototype combining KAG- and PAGED-inspired methods, with evaluation focused on procedural requirements, omissions, and structural correctness.
- Explored policy training for agentic RAG and trained colleagues on reinforcement learning with verifiable rewards, including ART and related tooling.
- Before moving into the CoE, contributed within a five-to-six-person team to end-to-end Python implementations of production RAG pipelines and evaluation, supporting their evolution toward agentic QA.

Lead Data Scientist / Senior Data Scientist

New York Life | 2017 - Dec 2022; Lead role Jan 2021 - Dec 2022

- Developed media mix models to support marketing budget allocation, with a 4-12% increase in weekly attributed sales.
- Developed spatial ML and LSTM travel-time models for agent operating areas, recruitment targeting, and site selection, including geospatial analysis across 2.5 million hexagons.
- Worked on speech analytics, transcription, and NLP initiatives across Agency and Marketing; helped establish reproducible ML development and coding practices.

EXPERIENCE, CONTINUED

Founding Machine Learning Engineer

Dejake | 2016 - 2017

- Developed neural sequence models and LSTM/RNN pipelines for e-commerce intent recognition, automated dialogue, and cart-recovery applications.

Assistant Director, Data Science - Solaria Labs

Liberty Mutual | 2013 - 2016

- Developed and deployed machine learning services for catastrophe severity modelling across Property and Auto portfolios, including event- and peril-level severity estimates.
- Built NLP feature-engineering services and sentiment-analysis pipelines for catastrophe-related policyholder text.

Machine Learning Software Engineer - SAS Enterprise Miner R&D

SAS Institute | 2007 - 2013

- Contributed algorithms and software to enterprise analytics products, spanning data mining, predictive modelling, and text-processing workflows.
- Implemented text-mining and statistical ML methods including TF-IDF, latent semantic analysis, rule-based parsing, and density-based clustering.

TECHNICAL EDUCATION & RECOGNITION

Colorless Green: Created a public retrieval course covering lexical and neural search, hybrid architectures, domain adaptation, and evaluation. colorlessgreen.ai

NLP & Machine Learning Instructor | Principal Analytics Prep., New York | Part-time, 2018 - 2020

Kaggle competition winner: Liberty Mutual Group - Fire Peril Loss Cost (2014).

EDUCATION

Graduate studies in Economics | University of Oxford & Louvain University

Advanced graduate coursework in Financial Economics.

B.S. in Economics | Carlos III University & Louvain University

SELECTED TECHNICAL SKILLS

Modelling & experimentation: PyTorch, Hugging Face Transformers, LoRA/QLoRA, domain-adaptive training, model comparison, ablation design, semantic analysis, retrieval and QA evaluation.

Retrieval & agent learning: BM25, dense retrieval, bi-encoders, cross-encoders, hybrid retrieval, knowledge graphs, ART, GRPO, reward and verification design.

Implementation: Python, SQL, Bash, AWS, Docker, Git, DVC, Weights & Biases, reproducible data and evaluation pipelines.